

Data Transformation Using Power BI

Removing Duplicates

The screenshot shows the Power BI Desktop interface with the 'Query Settings' pane open on the right. The 'APPLIED STEPS' list includes 'Source', 'Promoted Headers', 'Changed Type', 'Added Custom', 'Changed Type1', 'Added Conditional Column', 'Filtered Rows', and 'Removed Duplicates'. The 'Removed Duplicates' step is selected, indicating that duplicates have been removed from the data. The main data view shows a table with columns: Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The data is filtered to show only rows where the 'Order ID' is unique.

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1,275.00	-1148	7	Furniture	Bookcases
B-25602	168.00	-111	2	Electronics	Phones
B-25603	1,355.00	-60	5	Clothing	Trousers
B-25604	65.00	17	2	Clothing	T-shirt
B-25605	75.00	0	7	Clothing	Saree
B-25606	87.00	4	2	Clothing	Shirt
B-25607	50.00	15	4	Clothing	Leggings
B-25608	1,364.00	-1864	5	Furniture	Tables
B-25609	485.00	29	4	Electronics	Electronic Gs
B-25610	1,076.00	-38	4	Electronics	Printers
B-25611	160.00	-59	2	Clothing	Saree
B-25612	259.00	-55	2	Furniture	Chairs
B-25613	1,603.00	0	9	Clothing	Saree
B-25614	494.00	54	4	Furniture	Bookcases
B-25615	68.00	20	5	Clothing	Hankerchief
B-25616	42.00	12	5	Clothing	Hankerchief
B-25617	305.00	-270	5	Electronics	Electronic Gs
B-25618	362.00	127	1	Furniture	Bookcases
B-25619	353.00	90	8	Clothing	Saree
B-25620	193.00	46	1	Electronics	Printers
B-25621	233.00	-10	5	Electronics	Electronic Gs
B-25622	534.00	0	3	Clothing	Saree
B-25623	53.00	1	4	Clothing	Stole

Identifying the missing values in each column

The screenshot shows the Power BI Desktop interface with the 'Query Settings' pane open on the right. The 'Column profile' view is selected, displaying a bar chart for each column. The chart shows the distribution of values, including valid, error, and empty values. The 'Order ID' column has 500 distinct values and 500 unique values. The 'Amount' column has 300 distinct values and 198 unique values. The 'Profit' column has 227 distinct values and 144 unique values. The 'Quantity' column has 13 distinct values and 3 unique values. The 'Category' column has 3 distinct values and 0 unique values. The 'Sub-Category' column has 17 distinct values and 0 unique values.

Column	Valid	Error	Empty	Distinct	Unique
Order ID	100%	0%	0%	500	500
Amount	100%	0%	0%	300	198
Profit	100%	0%	0%	227	144
Quantity	100%	0%	0%	13	3
Category	100%	0%	0%	3	0
Sub-Category	100%	0%	0%	17	0

Sort by date value

The screenshot shows the Microsoft Power BI Desktop interface. The main view displays a table with the following columns: **itms**, **List of Orders.Order ID**, **List of Orders.Order Date**, **List of Orders.CustomerName**, and **List of Orders.State**. The data is sorted by **List of Orders.Order Date** in descending order. The table contains 24 rows of data, with dates ranging from 31-03-2019 to 22-03-2019. The right-hand pane shows the **Query Settings** for the selected query, with the **APPLIED STEPS** list showing **Sorted Rows**.

itms	List of Orders.Order ID	List of Orders.Order Date	List of Orders.CustomerName	List of Orders.State
1	B-26100	31-03-2019	Hitika	Madhya Pradesh
2	B-26099	30-03-2019	Bhishm	Maharashtra
3	B-26098	29-03-2019	Pinky	Jammu and Kashmir
4	B-26097	28-03-2019	Vini	Karnataka
5	B-26096	28-03-2019	Atharv	West Bengal
6	B-26095	28-03-2019	Monisha	Rajasthan
7	B-26094	27-03-2019	Deepak	Madhya Pradesh
8	B-26091	27-03-2019	Manju	Andhra Pradesh
9	B-26092	27-03-2019	Ramesh	Gujarat
10	B-26093	27-03-2019	Sarita	Maharashtra
11	B-26090	27-03-2019	Sagar	Nagaland
12	B-26088	26-03-2019	Bhavina	Sikkim
13	B-26086	26-03-2019	Mukesh	Haryana
14	B-26087	26-03-2019	Vandana	Himachal Pradesh
15	B-26085	26-03-2019	Shrichand	Punjab
16	B-26089	26-03-2019	Kanak	Goa
17	B-26084	25-03-2019	Anita	Kerala
18	B-26083	24-03-2019	Yogesh	Bihar
19	B-26082	23-03-2019	Jitesh	Uttar Pradesh
20	B-26078	22-03-2019	Kasheen	West Bengal
21	B-26077	22-03-2019	Divsha	Rajasthan
22	B-26080	22-03-2019	Sonakshi	Jammu and Kashmir
23	B-26079	22-03-2019	Hazel	Karnataka

Filter by state Tamil Nadu

The screenshot shows the Microsoft Power BI Desktop interface. The main view displays a table with the following columns: **itms**, **List of Orders.Order ID**, **List of Orders.Order Date**, **List of Orders.CustomerName**, and **List of Orders.State**. The data is filtered to show only rows where the **List of Orders.State** is **Tamil Nadu**. The table contains 8 rows of data, with dates ranging from 22-03-2019 to 08-04-2018. The right-hand pane shows the **Query Settings** for the selected query, with the **APPLIED STEPS** list showing **Filtered Rows**.

itms	List of Orders.Order ID	List of Orders.Order Date	List of Orders.CustomerName	List of Orders.State
1	B-26081	22-03-2019	Aarushi	Tamil Nadu
2	B-26018	14-02-2019	Aarushi	Tamil Nadu
3	B-26008	09-02-2019	Kalyani	Tamil Nadu
4	B-25860	15-11-2018	Akshay	Tamil Nadu
5	B-25788	21-09-2018	Dinesh	Tamil Nadu
6	B-25716	11-07-2018	Surabhi	Tamil Nadu
7	B-25698	23-06-2018	Amisha	Tamil Nadu
8	B-25608	08-04-2018	Aarushi	Tamil Nadu

Count of Order ID

The screenshot shows the Power BI Desktop interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The 'Transform' tab is active, showing various data transformation options. The query editor on the left shows a list of queries, with 'Order Details (3)' selected. The main area displays the query formula bar and a data preview table. The query formula is: `Table.Group(#"Filtered Rows", {"Order ID"}, {"Count", each Table.RowCount(_), Int64.Type})`. The data preview table has two columns: 'Order ID' and 'Count'. The 'Count' column is highlighted. The query settings pane on the right shows the query name 'Order Details (3)' and the applied steps: Source, Promoted Headers, Changed Type, Added Custom, Changed Type1, Added Conditional Column, Filtered Rows, and Grouped Rows.

Order ID	Count
B-25601	4
B-25602	5
B-25603	8
B-25604	2
B-25605	1
B-25606	1
B-25607	1
B-25608	4
B-25609	2
B-25610	6
B-25611	1
B-25612	1
B-25613	1
B-25614	2
B-25615	1
B-25616	4
B-25617	1
B-25618	2
B-25619	1
B-25620	1
B-25621	3
B-25622	1
B-25623	4
B-25624	1

Average By category

The screenshot shows the Power BI Desktop interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The 'Transform' tab is active, showing various data transformation options. The query editor on the left shows a list of queries, with 'Order Details (4)' selected. The main area displays the query formula bar and a data preview table. The query formula is: `Table.Group(#"Filtered Rows", {"Category"}, {"Average Profit", each List.Average([Profit]), type nullable})`. The data preview table has two columns: 'Category' and 'Average Profit'. The 'Average Profit' column is highlighted. The query settings pane on the right shows the query name 'Order Details (4)' and the applied steps: Source, Promoted Headers, Changed Type, Added Custom, Changed Type1, Added Conditional Column, Filtered Rows, and Grouped Rows.

Category	Average Profit
Furniture	9.456790123
Clothing	11.76290832
Electronics	34.07142857

Total target amount by month

The screenshot shows the Power BI Desktop interface. The 'Queries' pane on the left lists several queries, with 'Sales target (2)' selected. The main view displays a table with 12 rows and 2 columns: 'Month of Order Date' and '1.2 Total Target Amount'. The data represents monthly sales targets for the year 2018. The 'Query Settings' pane on the right shows the query name 'Sales target (2)' and the applied steps: 'Source', 'Promoted Headers', 'Changed Type', and 'Grouped Rows'.

Month of Order Date	1.2 Total Target Amount
01-04-2018	31400
01-05-2018	31500
01-06-2018	31600
01-07-2018	33800
01-08-2018	33900
01-09-2018	34000
01-10-2018	36100
01-11-2018	36300
01-12-2018	36400
01-01-2019	43500
01-02-2019	43600
01-03-2019	43800

Total amount by sub category

The screenshot shows the Power BI Desktop interface. The 'Queries' pane on the left lists several queries, with 'Order Details (5)' selected. The main view displays a table with 17 rows and 2 columns: 'Sub-Category' and '1.2 Total Amount'. The data represents the total amount for various sub-categories. The 'Query Settings' pane on the right shows the query name 'Order Details (5)' and the applied steps: 'Source', 'Promoted Headers', 'Changed Type', 'Added Custom', 'Changed Type1', 'Added Conditional Column', 'Filtered Rows', and 'Grouped Rows'.

Sub-Category	1.2 Total Amount
Bookcases	56861
Stole	18546
Hankierchief	14608
Electronic Games	39168
Phones	46119
Saree	53511
Trousers	30039
Chairs	34222
Kurti	3361
T-shirt	7382
Shirt	7555
Leggings	2106
Tables	22614
Printers	58252
Accessories	21728
Furnishings	13484
Skirt	1946

Relationship between Sales, Order Details & order data

